Results on PBM-driven ODE discovery and ODE parameter estimation

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ODE discovery with Probmot

- processes between entities (predator-prey models,...)
- build a list of possible models $[\vec{x}'(t) = \vec{F}_i(\vec{x}(t), t, \theta^{(i)})]_i$
- **estimate model parameters** $\theta^{(i)}$
- evaluate all possible models

Structure changes

- migration from Java to Python
- gradient opt instead of EA
- magnitutes faster parameter estimation
- conditional statemets support

$$x'(t) = \begin{cases} f(x,t,\theta) & \phi(x,t,\theta) > 0 \\ g(x,t,\theta) & \text{otherwise} \end{cases}$$

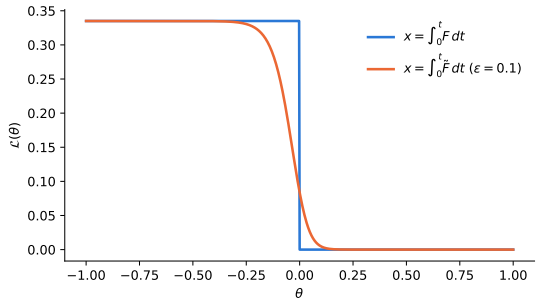
Other possible solutions

- JIT compiled $F(x, t, \theta)$
- clever structure search
- precise estimation just for best candidates

Relaxed conditionals

$$F = \begin{cases} f & \theta \geq 0 \\ g & \text{otherwise} \end{cases}$$

$$\tilde{F} = \frac{f+g}{2} + \frac{f-g}{2} \tanh \frac{\theta}{\varepsilon}$$



Parameter estimation

- $x\colon [t_0, t_1] \to \mathbb{R}^n$
- $x_{\text{obs}} = (x_{o_1}, x_{o_2}, \ldots, x_{o_{n_{\text{obs}}}}) \in \mathbb{R}^{n_{\text{obs}}}$

Given

- ODE $x'(t) = F(x(t), t, \theta)$
- measurments $(t_i, x_{\text{obs}}(t_i) + \epsilon_i)$, $\epsilon_i \sim N(\mu, \sigma^2)$

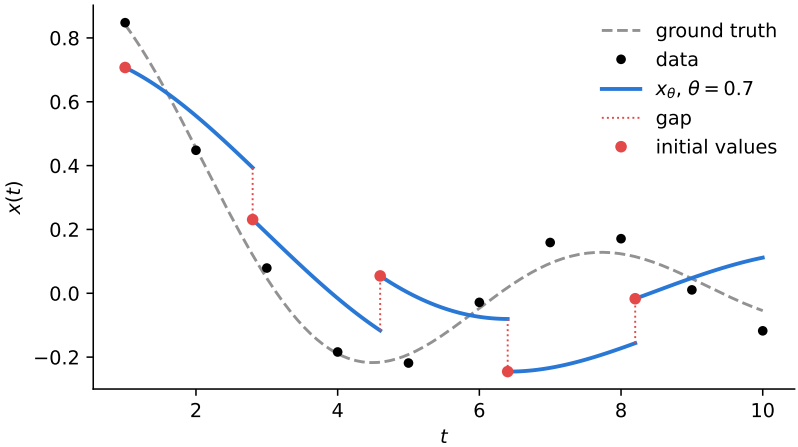
estimate θ.

Integration using multishooting[1]

- integration: pick $x(t_0)$, solve ODE for $x_\theta(t)$, min discrepancy $\mathcal{L}(x_\theta)$ on data
- problem: caotic system
- solution: integrate shorter intervals, join trajectories

[1] Ozgur Aydogmus and Ali Hakan Tor (2020). *A Modified Multiple Shooting Algorithm for Parameter Estimation in ODEs Using Adjoint Sensitivity Analysis*. arXiv: 2006.15740 [math.OC].

Integration using multishooting



Multiple shooting for $x(t) = \sin(\theta t)/t$ on $[1, 5]$, $\theta_{\text{true}} = 1$, evaluated at $\theta = 0.7$ with 10 intervals. Data: 16 points with noise $\mathcal{N}(0, 0.05^2)$; initial values perturbed by $\mathcal{U}(-0.1, 0.1)$.

Integration using multishooting[2]

Multishooting

minimize discrepancy on data

$$\sum_i \mathcal{L}(x_i), \quad \text{where } x_i' = F(x_i, t, \theta), x_i(t_i) = s_i$$

on

$$x_i(t_{i+1}) = s_i \quad \forall i = 0, 1, \ldots N_{\text{subintervals}}$$

[2] Ozgur Aydogmus and Ali Hakan Tor (2020). *A Modified Multiple Shooting Algorithm for Parameter Estimation in ODEs Using Adjoint Sensitivity Analysis.* arXiv: 2006.15740 [math.OC].

Matching

Integral matching

Batteling caotic loss surface

Integral lenght, homotopic methods

Collocation

matching on unobserved

Note on MAGI

TODO razlike

Question: lenght of integration interval

Results

Back to ProBmot